MID TERM EXAMINATION

Paper: Mid Term Examination - Set B | Duration: 3 Hours | Max Marks: 53 | Date: 14-03-2026

Total Questions: 15 | Answer all questions.

Name: ____________________________ **Roll No:** _________________________

Student Sign: ___________________ **Teacher Sign:** ___________________

SECTION A: MCQS

1. Which of the following operations is NOT typically associated with a standard Queue data structure? **[2 Marks]**

a) Enqueue
b) Dequeue
c) Pop
d) Front

2. In a circular linked list, what does the 'next' pointer of the last node point to? **[2 Marks]**

a) NULL
b) Itself
c) The first node of the list
d) The second node of the list

3. What is the main structural difference between an array and a record (struct/object) in terms of the types of data they can store? **[2 Marks]**

SECTION A: SHORT ANSWERS

4. What is the primary characteristic that distinguishes a linear data structure from a non-linear data structure? **[2 Marks]**

5. Which two primary operations define the behavior of a Queue data structure? **[2 Marks]**

SECTION A: MCQS

6. What is an Abstract Data Type (ADT)? **[3 Marks]**

a) A specific implementation of a data structure in a programming language.
b) A mathematical model that defines the data values and the operations on that data, without specifying their implementation.
c) A data structure that stores only abstract values or pointers.
d) A pre-defined class or library in a programming language.

7. If an algorithm has a time complexity of O(n^2), and its input size n is doubled, how does its execution time approximately change in the worst case? **[3 Marks]**

a) It doubles.

b) It quadruples.
c) It remains the same.
d) It halves.

8. When implementing a stack using a singly linked list, which end should new elements be pushed onto for O(1) efficiency, assuming head points to the top of the stack? **[3 Marks]**

a) The tail end of the list.
b) The middle of the list.
c) The head end of the list.
d) It does not matter which end, as all operations are O(1).

SECTION A: SHORT ANSWERS

9. In algorithm analysis, what does it signify if an algorithm has an O(1) time complexity? **[3 Marks]**

10. A programmer needs to design a system where customer support requests are handled based on their severity level (e.g., critical, high, medium, low). Which variant of a queue would be most suitable for this scenario? **[3 Marks]**

11. Briefly explain the concept of Space Complexity in algorithm analysis. What are the two main types of space that contribute to it? **[5 Marks]**

12. Discuss two common applications where a Queue data structure is primarily used. **[5 Marks]**

SECTION A: MCQS

13. Describe the steps to delete a node with a specific value X from a singly linked list, assuming head is the pointer to the first node. **[6 Marks]**

14. Using a linear array-based implementation of a queue, describe how you would determine if the queue is full and if it is empty, assuming front and rear pointers are used and MAX_SIZE is the maximum capacity of the array. **[6 Marks]**

SECTION A: SHORT ANSWERS

15. Explain the difference between "Pass by Value" and "Pass by Reference" when passing a singly linked list (represented by its head pointer) to a function. Discuss the implications of each method on whether the function can modify the original list structure (e.g., change the head). **[6 Marks]**

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